

A genus obstruction for commuting boundary operators

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Abstract

Let M be a compact connected oriented smooth Riemannian surface with nonempty smooth boundary. We prove that if its Dirichlet-to-Neumann map commutes with the boundary Laplacian, then M has genus zero. This answers both cases of Speciel's Open Problem 1.6 and, together with his genus-zero classification, removes the topological restriction from the classification of oriented surfaces with commuting boundary operators. For connected boundary, the finite-dimensional obstruction to harmonic conjugation forces an exact determinant identity on all sufficiently high Fourier planes. The algebra of holomorphic boundary traces then produces a holomorphic map of degree one. For several boundary components, we prove that conformally attaching a Euclidean disk along an arclength identification preserves commutation. The proof separates the finite-period input, the multiplicative trace argument, and the variational elimination of the capped boundary.

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1. Introduction

Let (M, g) be a compact connected smooth Riemannian surface with nonempty smooth boundary. Its Dirichlet-to-Neumann map is

$$Af = \partial_\nu u_f|_{\partial M}, \quad \Delta_g u_f = 0, \quad u_f|_{\partial M} = f, \quad (1.1)$$

where ν is the outward unit normal and Δ_g is the nonnegative Laplacian. Write Δ_∂ for the nonnegative Laplacian of the induced boundary metric. We consider the exact commutation condition

$$A\Delta_\partial f = \Delta_\partial Af \quad \text{for every } f \in C^\infty(\partial M). \quad (1.2)$$

Speciel [S, Theorem 1.3] classified the surfaces satisfying (1.2) under the additional assumption that the genus is zero or that there are at least three boundary components. His classification contains disks, logarithmic ovals, and flat cylinders, with the equivalence appropriate to two-dimensional Dirichlet-to-Neumann data. The remaining cases, a positive-genus oriented surface with one or two boundary components, are explicitly posed in [S, Open Problem 1.6]. The numbering here refers to the final journal article; the corresponding question is Open Problem 1.12 in its arXiv version.

Theorem 1.1 (Genus obstruction). Let (M, g) be a compact connected oriented smooth Riemannian surface with nonempty smooth boundary. If (1.2) holds, then M has genus zero.

The theorem imposes no curvature condition and no restriction on the number or lengths of the boundary components. In particular, neither of the positive-genus surfaces in [S, Open Problem 1.6] admits the requested metric.

Recall that two surfaces are σ -isometric if a diffeomorphism identifies their metrics after a conformal change $g \mapsto e^{2\varphi}g$ with $\varphi = 0$ on the boundary. Both operators in (1.2) are preserved by this equivalence. A logarithmic oval is, up to Euclidean similarities, the image of the closed unit disk under $z \mapsto \log(z-a)$, $a > 1$, using a branch holomorphic near the disk. We use the flat cylinder models of [S].

Corollary 1.2 (Classification without a genus restriction). A compact connected oriented smooth Riemannian surface with nonempty smooth boundary satisfies (1.2) if and only if it is σ -isometric to a disk, a logarithmic oval, or a flat cylinder.

Proof. Theorem 1.1 reduces the assertion to the genus-zero case of [S, Theorem 1.3], which supplies both the classification and its converse. $\square$

The new assertion is the genus obstruction. The models, their commutation property, and their classification in genus zero belong to Speciel. Our proof does not start from a uniformization of a positive-genus surface by a planar domain. Instead, it constructs a degree-one holomorphic map from the boundary operator identity.

When the boundary is connected, let T denote positively oriented arclength differentiation. Harmonic conjugation can fail globally because of periods, but these periods lie in a finite-dimensional cohomology space. Consequently, on mean-zero functions, $(A^{-1}T)^2 + I$ has finite rank. This is a classical feature of the boundary Hilbert transform in the inverse problem for Riemann surfaces; see [B] and the formulation in [BK, Section 2]. We give the needed period argument directly. Under (1.2), the finite-rank defect becomes an exact determinant constraint on every sufficiently high two-dimensional Fourier plane.

The further input is multiplicative. Smooth holomorphic boundary traces form an algebra, characterized by $Af = -iTf$. Condition (1.2) makes that algebra invariant under Fourier-plane projections. Multiplying adjacent high-frequency traces and projecting to frequency one constructs an ellipse-valued boundary parametrization. The exceptional case in which all negative high-frequency coefficients vanish produces a disk-valued parametrization by taking a quotient of two holomorphic functions. In both cases, the argument principle proves that the map has degree one.

For disconnected boundary we attach a Euclidean disk along one boundary component by smooth conformal welding. Two-dimensional Dirichlet energy is unchanged on each piece by conformal rescaling. Minimizing over the seam trace gives a Schur complement of the original Dirichlet-to-Neumann map and the disk operator. The boundary arclength identification makes the latter the square root of the seam Laplacian. This proves that commutation survives capping, including at the constant mode. Iterating reduces to connected boundary without changing genus.

2. Holomorphic traces and the period defect

We use the complex structure determined by the orientation and conformal class of g . A holomorphic function on M will mean one holomorphic in the interior and smooth up to the boundary. Orient ∂M positively, so that (ν, τ) is a positively oriented orthonormal frame when τ is the positive unit tangent. Set $T = \partial_\tau$ on each component. Our Hodge-star convention is $*dx = dy$ in an oriented isothermal coordinate $x + iy$.

For complex-valued traces we extend A complex linearly. Green's identity gives

$$\int_{\partial M} \bar{f} A f ds = \int_M (|du|^2 + |dv|^2) dV_g, \quad u + iv = u_f. \quad (2.1)$$

In particular, A is self-adjoint and nonnegative on $L^2(\partial M, ds)$; its kernel consists of the global constants, since M is connected. Smooth elliptic boundary regularity is understood throughout.

Lemma 2.1 (Holomorphic trace criterion). A function $f \in C^\infty(\partial M; \mathbb{C})$ is the trace of a holomorphic function on M if and only if

$$A f = -i T f. \quad (2.2)$$

Proof. If $U = u + iv$ is holomorphic, then $dv = *du$. At the boundary this gives $Tv = Au$ and $Av = -Tu$, proving (2.2).

Conversely, extend f harmonically to $U = u + iv$. By (2.1), (2.2), and Stokes' theorem,

$$\begin{aligned} \int_M (|du|^2 + |dv|^2) dV_g &= -i \int_{\partial M} \bar{f} df \\ &= \int_{\partial M} (u dv - v du) = 2 \int_M du \wedge dv. \end{aligned} \quad (2.3)$$

The imaginary term is $-i \int_{\partial M} d(u^2 + v^2)/2 = 0$. Pointwise,

$$|dv - *du|^2 dV_g = (|du|^2 + |dv|^2) dV_g - 2du \wedge dv. \quad (2.4)$$

Its integral is zero by (2.3), hence $dv = *du$. Thus U is holomorphic. $\square$

Let $\mathcal{K}$ be the space of these boundary traces. It contains the constants and is closed under multiplication. Lemma 2.1 identifies it as the kernel of $A + iT$ on smooth complex functions. This standard trace-algebra viewpoint is also used in the inverse problem; our use below concerns its additional Fourier invariance under (1.2).

From now through Section 3, assume that ∂M is connected. A constant rescaling of g preserves (1.2), since A and Δ_∂ scale with degrees -1 and -2 . Normalize its length to 2π . Positively oriented arclength is then $s \in \mathbb{R}/2\pi\mathbb{Z}$, and $T = d/ds$.

Let $C_0^\infty(\partial M; \mathbb{R})$ denote the real smooth functions with zero boundary mean. The restriction of A to this space has a smooth inverse: the Neumann problem is solvable precisely for mean-zero normal data, and its additive constant is fixed by the boundary mean. Define

$$H = A^{-1}T : C_0^\infty(\partial M; \mathbb{R}) \longrightarrow C_0^\infty(\partial M; \mathbb{R}). \quad (2.5)$$

Lemma 2.2 (Finite period defect). If M has genus γ and one boundary component, then $H^2 + I$ has rank at most 2γ on $C_0^\infty(\partial M; \mathbb{R})$.

Proof. If u is the harmonic extension of f , then $*du$ is closed. Consider the linear period map

$$\mathcal{P} : f \longmapsto [*du] \in H_{\text{dR}}^1(M; \mathbb{R}). \quad (2.6)$$

This cohomology space has dimension 2γ . If $\mathcal{P}f = 0$, there is a smooth real function v with $dv = *du$. It is harmonic, and we choose its additive constant so that its boundary trace has mean zero. The boundary Cauchy-Riemann identities are

$$A(v|_{\partial M}) = -Tf, \quad T(v|_{\partial M}) = Af. \quad (2.7)$$

Consequently $Hf = -v|_{\partial M}$ and

$$H^2f = -A^{-1}T(v|_{\partial M}) = -f. \quad (2.8)$$

Thus $H^2 + I$ vanishes on $\ker \mathcal{P}$ and factors through a vector space of dimension at most 2γ . This proves the rank bound. No existence of a harmonic conjugate is asserted outside $\ker \mathcal{P}$. $\square$

The exact topological rank formula for a related boundary Hilbert transform is stronger than Lemma 2.2; see [BK, Section 2, formulas (12)-(16)] and its attribution to [B, Lemma 1]. Only the elementary upper bound just proved is needed here.

3. Connected boundary forces genus zero

Assume (1.2), and keep the normalization $|\partial M| = 2\pi$. For $n \geq 1$, let

$$E_n = \text{span}_{\mathbb{R}}\{\cos(ns), \sin(ns)\}. \quad (3.1)$$

These are the positive eigenspaces of $\Delta_{\partial} = -T^2$. Exact commutation implies that A preserves each E_n . Self-adjointness then implies that its orthogonal Fourier projections commute with A on smooth functions. The operators A^{-1} , T , and H also preserve E_n .

Lemma 3.1 (High Fourier planes). There exists $N \geq 1$ such that for every $n \geq N$ there is a unique $c_n \in \mathbb{C}$, $|c_n| < 1$, for which

$$f_n(s) = e^{ins} + c_n e^{-ins} \in \mathcal{K}. \quad (3.2)$$

Proof. In the real orthonormal cosine-sine basis of E_n , write A_n for the positive symmetric matrix of A and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The matrix of T is nJ . For any positive symmetric two-by-two matrix S , direct multiplication gives

$$(S^{-1}J)^2 = -\frac{1}{\det S}I. \quad (3.3)$$

It follows that

$$(H^2 + I)|_{E_n} = \left(1 - \frac{n^2}{\det A_n}\right)I. \quad (3.4)$$

If the scalar in (3.4) were nonzero for infinitely many n , the range of $H^2 + I$ would contain infinitely many independent Fourier planes. This contradicts Lemma 2.2. Hence

$$\det A_n = n^2 \quad (n \geq N) \quad (3.5)$$

for some N .

Set $z = e^{is}$. In the complex orthonormal basis proportional to (z^n, z^{-n}) , reality and self-adjointness give

$$A_n = \begin{pmatrix} a & b \\ \bar{b} & a \end{pmatrix}, \quad a > 0, \quad -iT = \begin{pmatrix} n & 0 \\ 0 & -n \end{pmatrix}. \quad (3.6)$$

By (3.5), $\det(A_n + iT) = a^2 - |b|^2 - n^2 = 0$. Its kernel is one-dimensional, because its lower diagonal entry is $a + n > 0$. Lemma 2.1 identifies this kernel with $\mathcal{K} \cap E_n^{\mathbb{C}}$.

For any nonzero trace $f = \alpha z^n + \beta z^{-n}$ in this kernel, strict positivity of (2.1) and (2.2) imply

$$0 < \int_{\partial M} \bar{f} A f \, ds = 2\pi n(|\alpha|^2 - |\beta|^2). \quad (3.7)$$

The trace is nonconstant, so its harmonic extension has strictly positive energy. Thus $\alpha \neq 0$ and $|\beta/\alpha| < 1$. Normalization proves existence and uniqueness in (3.2). $\square$

Write P_n for the orthogonal projection onto $E_n^{\mathbb{C}}$. Since both A and T commute with P_n , Lemma 2.1 gives

$$P_n \mathcal{K} \subset \mathcal{K} \quad (n \geq 1). \quad (3.8)$$

Proposition 3.2 (A degree-one boundary trace). There is a holomorphic function F on M whose boundary trace has the form

$$F|_{\partial M} = z + cz^{-1}, \quad |c| < 1. \quad (3.9)$$

Proof. First suppose that $c_n \neq 0$ for some $n \geq N$. The product $f_n f_{n+1}$ belongs to $\mathcal{K}$. Its frequency-one projection is

$$P_1(f_n f_{n+1}) = c_n z + c_{n+1} z^{-1}. \quad (3.10)$$

This is a nonzero holomorphic trace by (3.8). Applying (3.7) at frequency one shows that $|c_{n+1}| < |c_n|$. Dividing (3.10) by c_n proves (3.9).

Suppose instead that $c_n = 0$ for all $n \geq N$. Fix such an n . Let F_n, F_{n+1} be the holomorphic extensions of z^n, z^{n+1} . The holomorphic function $F_n^{n+1} - F_{n+1}^n$ has zero boundary trace, so the maximum principle gives

$$F_n^{n+1} = F_{n+1}^n \quad \text{on } M. \quad (3.11)$$

The quotient $G = F_{n+1}/F_n$ is meromorphic in the interior, is smooth near the boundary since F_n is nonzero there, and satisfies $G^n = F_n$ wherever it is initially defined. A pole of G would give a pole of G^n , which is impossible. Its apparent singularities are therefore removable. Thus G is holomorphic on M , smooth to the boundary, and has boundary trace z . Take $F = G$ and $c = 0$. $\square$

Proposition 3.3 (Connected-boundary case of Theorem 1.1). If ∂M is connected and (1.2) holds, then M has genus zero.

Proof. Take F from Proposition 3.2. The real-linear map $w \mapsto w + c\bar{w}$ has determinant $1 - |c|^2 > 0$. Hence (3.9) maps the oriented boundary diffeomorphically onto a positively oriented nondegenerate ellipse Γ . Let Ω be the bounded component of $\mathbb{C} \setminus \Gamma$.

For $a \notin \Gamma$, the argument principle on the bordered Riemann surface gives

$$\sum_{p \in F^{-1}(a)} \text{ord}_p(F - a) = \frac{1}{2\pi i} \int_{\partial M} \frac{dF}{F - a} = \begin{cases} 1, & a \in \Omega, \\ 0, & a \notin \overline{\Omega}. \end{cases} \quad (3.12)$$

There are finitely many zeros in (3.12), since there are none near the boundary. For completeness, the first equality follows by deleting small disjoint coordinate disks around those zeros and applying Stokes' theorem to the closed form $dF/(F - a)$ on the remaining surface. Each small circle contributes its positive zero multiplicity.

No interior point can map to Γ : by the open mapping theorem its image neighborhood would contain points outside $\overline{\Omega}$, contradicting the second case of (3.12). Therefore F maps the interior bijectively onto Ω , and every local multiplicity is one. It is a biholomorphism there. Its boundary map is also bijective, so $F : M \rightarrow \overline{\Omega}$ is a homeomorphism by compactness. In particular M is a topological disk and has genus zero. $\square$

4. Conformal capping preserves commutation

We now allow several boundary components. The following construction concerns Dirichlet energy and keeps the line element on every uncapped component exactly fixed.

Lemma 4.1 (Smooth conformal cap). Suppose M has at least two boundary components, and designate one of them by Y . Let D be a Euclidean disk of perimeter $|Y|_g$. Identify ∂D with Y by an orientation-reversing arclength isometry. The resulting topological surface admits a smooth conformal structure and a smooth compatible metric $\widehat{g}$ with these properties:

1. The inclusions of M and D are conformal on their interiors and smooth with nondegenerate derivatives up to the seam.
2. The line element of $\widehat{g}$ on $X = \partial M \setminus Y$ equals the original line element of g .
3. If $U \in H^1(\widehat{M})$ has restrictions u and v to the two pieces, then

$$\int_{\widehat{M}} |dU|_{\widehat{g}}^2 dV_{\widehat{g}} = \int_M |du|_g^2 dV_g + \int_D |dv|^2 dx. \quad (4.1)$$

The surface $\widehat{M}$ is connected, has one fewer boundary component, and has the same genus as M .

Proof. Smooth conformal sewing along an orientation-reversing boundary diffeomorphism is standard; a general bordered-surface statement is [T, Theorem 2.33]. The precise local smoothness can also be obtained from disk welding [HT, Theorem 3.1] and disk-annulus welding [HT, Proposition 3.6]. Here is the reduction to that local form.

Choose a smooth annular collar of Y in M . Its conformal uniformization by a round annulus is smooth with nonzero derivative at both boundary circles. In these coordinates the seam identification is a smooth circle diffeomorphism. Welding the disk to the collar produces a disk with conformal inclusions smooth up to a smooth seam. The conformal structure on the open interior of the original collar is unchanged. It therefore glues to the remainder of M along an open collar overlap. This supplies the asserted smooth conformal structure on $\widehat{M}$.

Choose any smooth metric h compatible with this structure. Its restriction to M is conformal to g , and its restriction to D is conformal to the Euclidean metric. On each component of X , prescribe a positive smooth scalar factor so that a conformal rescaling of h has the original line element. Extend this factor smoothly and positively to $\widehat{M}$, using boundary collars, and denote the rescaled metric by $\widehat{g}$.

In dimension two, $|du|_{e^{2\varphi}g}^2 dV_{e^{2\varphi}g} = |du|_g^2 dV_g$. Applying this on both pieces proves (4.1). Matching $H^{1/2}$ traces across the smooth seam are precisely the compatibility condition for two piecewise H^1 functions to define an H^1 function on $\widehat{M}$. This follows in a smooth seam chart from the usual trace and gluing property for a hypersurface; a jump is the only additional distributional derivative. No matching of normal metric derivatives across the seam is required. Finally, attaching a disk increases Euler characteristic by one and decreases the boundary count by one, so the genus is unchanged. $\square$

The arclength condition determines the disk operator transported to the seam:

$$B = \Delta_Y^{1/2}. \quad (4.2)$$

Indeed, on a disk of radius $r = |Y|_g/(2\pi)$, the modes $e^{in\theta}$ have Dirichlet-to-Neumann eigenvalues $|n|/r$ and boundary Laplace eigenvalues n^2/r^2 . Reversing orientation leaves both operators unchanged.

Relative to $L^2(X) \oplus L^2(Y)$ with the original line elements, write the original operator in blocks as

$$A = \begin{pmatrix} A_{XX} & A_{XY} \\ A_{YX} & A_{YY} \end{pmatrix}. \quad (4.3)$$

Proposition 4.2 (Capping and commutation). If A commutes with Δ_∂ , then the Dirichlet-to-Neumann map $\widehat{A}$ of $(\widehat{M}, \widehat{g})$ in Lemma 4.1 commutes with the unchanged boundary Laplacian Δ_X .

Proof. Let d_M and $d_{\widehat{M}}$ denote the minimized Dirichlet-energy forms for prescribed boundary values. Formula (4.1) and the trace gluing property give, for $f \in H^{1/2}(X)$,

$$d_{\widehat{M}}[f] = \inf_{h \in H^{1/2}(Y)} \{d_M[(f, h)] + \langle Bh, h \rangle\}. \quad (4.4)$$

The brackets in (4.4) denote Sobolev duality, agreeing with the L^2 energy pairing on smooth real functions; complex forms follow by polarization. Minimization over the interior of each piece first produces its harmonic extension, which justifies both directions of (4.4).

The seam operator

$$C = A_{YY} + B : H^{1/2}(Y) \longrightarrow H^{-1/2}(Y) \quad (4.5)$$

is coercive. In fact, if u_h has boundary trace $(0, h)$ on $X \sqcup Y$, then the Poincaré inequality with zero trace on the nonempty boundary portion X , followed by the trace theorem, yields

$$\|h\|_{H^{1/2}(Y)}^2 \leq C_0 \|u_h\|_{H^1(M)}^2 \leq C_1 \int_M |du_h|^2 dV_g = C_1 \langle A_{YY} h, h \rangle. \quad (4.6)$$

Adding the nonnegative disk energy preserves coercivity. The bounded form of C is therefore invertible by the variational existence theorem. This includes constants on Y : although B annihilates constants, the extension with zero trace on X supplies the positive term in (4.6).

The unique minimizing trace in (4.4) satisfies

$$Ch = -A_{YX} f. \quad (4.7)$$

Polarization, or differentiation of the minimum energy with respect to f , gives the exact Schur complement

$$\hat{A} = A_{XX} - A_{XY}C^{-1}A_{YX}. \quad (4.8)$$

The boundary measure used here is the original measure on X , which equals the one for $\hat{g}$ by Lemma 4.1. Thus (4.8) is the geometric Dirichlet-to-Neumann map, with no additional boundary multiplier.

It remains to check commutation without an unjustified manipulation of unbounded operators. The original commutation relation gives the block identities

$$\Delta_i A_{ij} = A_{ij} \Delta_j, \quad i, j \in \{X, Y\}, \quad (4.9)$$

on smooth functions. The diagonal block A_{YY} is self-adjoint and preserves every Δ_Y eigenspace, so it commutes with its orthogonal spectral projections. The same is true of B and hence of the form operator C . These projection identities extend to the indicated Sobolev spaces by density. Its inverse also commutes with the projections: apply a projection to $Ch = k$ and use uniqueness of the solution.

Now let f belong to the λ eigenspace of Δ_X . By (4.9), $A_{YX}f$ belongs to the λ eigenspace of Δ_Y , or is zero if that eigenvalue is absent there. Equation (4.7) therefore puts h in the same eigenspace. Equations (4.8)-(4.9) imply that $\hat{A}f$ is in the λ eigenspace of Δ_X . This also holds for $\lambda = 0$ by the invertibility already proved. Hence $\hat{A}$ preserves every Δ_X eigenspace. Smooth Fourier expansion on the finite union of boundary circles, together with continuity of both operators on smooth functions, proves their commutation. $\square$

Proof of Theorem 1.1. If the boundary is connected, use Proposition 3.3. Otherwise cap one boundary component using Lemma 4.1 and Proposition 4.2. The resulting surface has the same genus, one fewer boundary component, and commuting boundary operators. Repeat until the boundary is connected. Proposition 3.3 makes the final genus zero, hence the original genus was zero. $\square$

5. Scope and further questions

The proof uses exact commutation twice: first to reduce the finite-period defect to scalar defects on Fourier planes, and then to preserve the holomorphic-trace algebra under projection. A small commutator does not imply either exact determinant identities or exact multiplicative Fourier relations. A quantitative genus obstruction would require additional geometric control and is not a consequence of this argument. The stability results in [S2] concern other settings and do not supply that conclusion for arbitrary bordered surfaces.

Proposition 4.2 may be useful separately when studying boundary symmetries compatible with conformal sewing. Its hypotheses specify the metric on the uncapped boundary and the arclength identification at the seam; omitting either can change the operators in (4.8). The present theorem concerns orientable surfaces with smooth metrics and smooth boundary. We make no classification claim for nonorientable surfaces or for rough boundaries.

Research and preparation statement. This manuscript was developed and written with AI assistance, including its proof development, source comparison, and internal audit. It has not received independent expert review. The accompanying source and review records distinguish prior results, the proposed contribution, and the limits of the priority search. No numerical computation is used as evidence for the theorem.

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